Basic Mathematical Concepts

Cryptography

RECAP: Computer characters

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Cryptography

Computer characters

A => 65
B => 66
C => 67
D => 68
E => …

a => 97
b => 98
c => 99
d => 100
e => …

characters and their decimal numbers

Cryptography

Computer characters

Computers communicate in bits.

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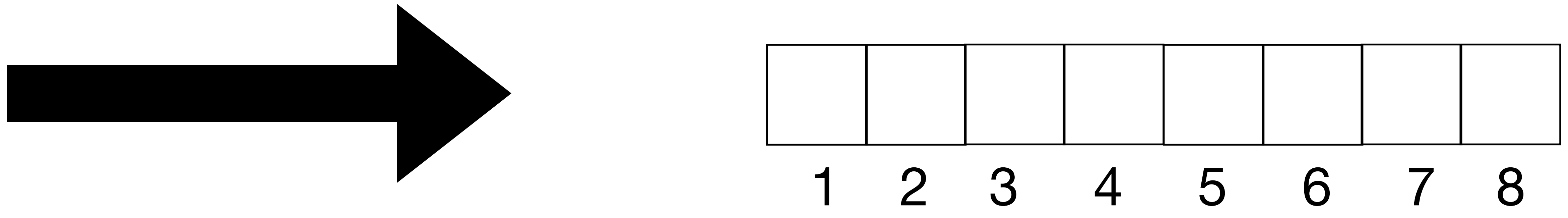
characters and their decimal numbers



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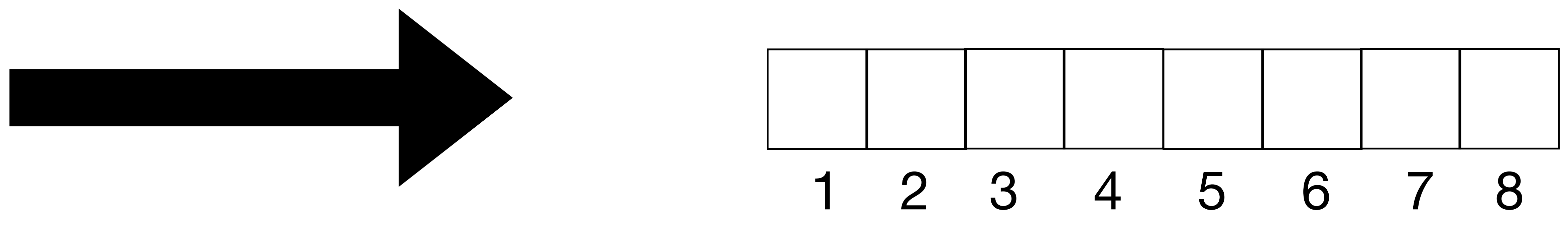

A => 0100 0001
B => 0100 0010
C => 0100 0011
D => 0100 0100
E => …

a => 0110 0001
b => 0110 0010
c => 0110 0011
d => 0110 0100
e => …

characters and their binary numbers

Cryptography

Recap: Decimal to Binary

A => 65	a => 97
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characters and their decimal numbers

Conversion steps:

1. **Divide the number by 2.**
2. **Get the integer quotient for the next iteration.**
3. **Get the remainder for the binary digit.**
4. **Repeat the steps until the quotient is equal to 0.**

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Example: Convert 15 to binary

$15 \div 2 = 7$ (Remainder 1)
$7 \div 2 = 3$ (Remainder 1)
$3 \div 2 = 1$ (Remainder 1)
$1 \div 2 = 0$ (Remainder 1)

0	0	0	0	1	1	1	1
1	2	3	4	5	6	7	8

Cryptography

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0	0	0	0	1	1	1	1
1	2	3	4	5	6	7	8

Example: Convert 8 to binary

8 ÷ 2 = 4 (Remainder 0)
4 ÷ 2 = 2 (Remainder 0)
2 ÷ 2 = 1 (Remainder 0)
1 ÷ 2 = 0 (Remainder 1)

0	0	0	0	1	0	0	0
1	2	3	4	5	6	7	8

Cryptography

Recap: Decimal to Binary

Covert the following decimals to binary

- **16**
- **22**
- **11**
- **64**
- **255**

Cryptography

Recap: Binary to Decimal

For a binary with n digits

$$d_{n-1} \ldots d_3 \; d_2 \; d_1 \; d_0$$

The decimal is equal to the sum of the binary digits (d_n) times their power of 2^n

$$\textbf{Decimal} = d_0 \times 2^0 + d_1 \times 2^1 + d_2 \times 2^2 + \ldots$$

Cryptography

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Example

Binary number	0	0	0	0	0	1	0	1
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

$$\text{Decimal} = 0\times2^7 + 0\times2^6 + 0\times2^5 + 0\times2^4 + 0\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$$
$$= 5$$

Cryptography

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$$= 5$$

Cryptography

Recap: Binary to Decimal

Covert the following binary to Decimal

- **0000 0010**
- **0001 0011**
- **0001 0111**
- **0101 0110**
- **0000 0001**

Cryptography

Recap: LOGIC GATES

AND Truth Table

Inputs		Output
A	B	Y = A.B
0	0	0
0	1	0
1	0	0
1	1	1

OR Truth Table

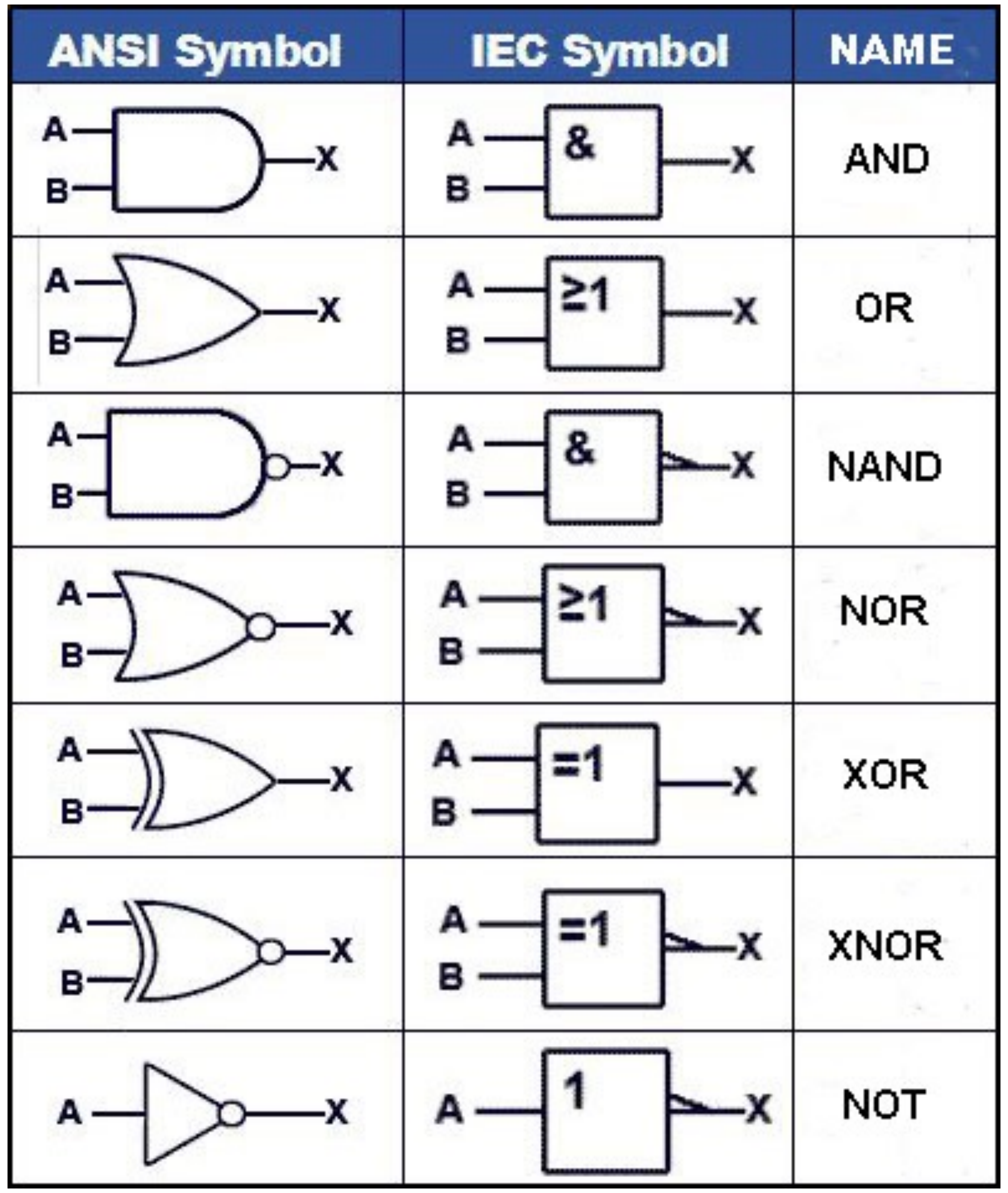
Inputs		Output
A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

NAND Truth Table

Inputs		Output
A	B	$Y = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

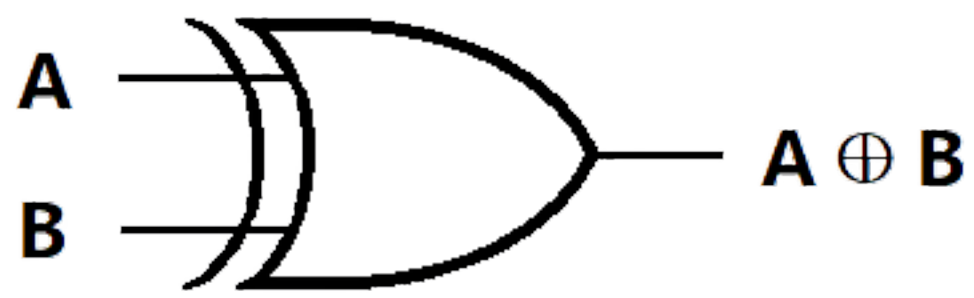
XOR Truth Table

Inputs		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0



Cryptography

Recap: LOGIC GATES



XOR ↓

0 0 0 1 0 1 1 1

0 1 0 1 0 1 0 1

AND Truth Table

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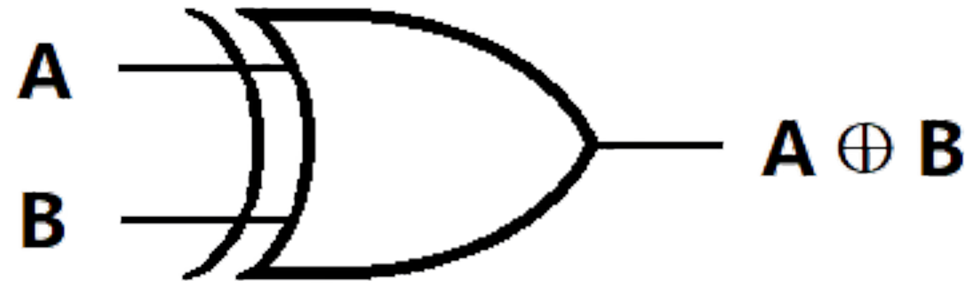

XOR ↓

0 0 0 1 0 1 1 1

0 1 0 1 0 1 0 1

$\boxed{0\,1\,0\,0\quad 0\,0\,1\,0}$

Cryptography

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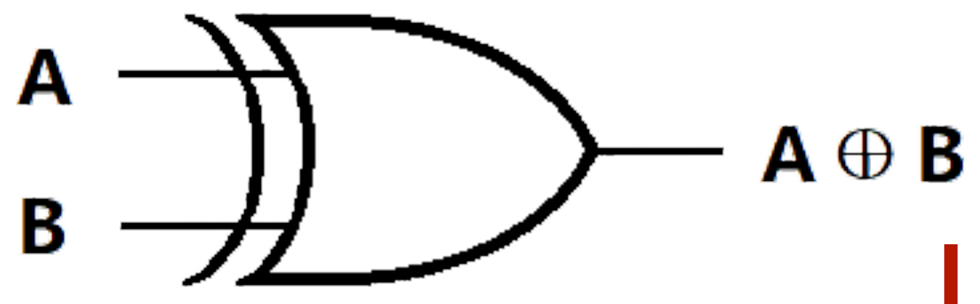

XOR ↓

0 0 0 1 0 1 1 1

0 1 0 1 0 1 0 1

$\boxed{\textbf{0 1 0 0 0 0 1 0}}$

Can you XOR the following binary

(A.)

1 0 1 1 0 0 1 1

0 1 0 0 1 1 0 1

(B.)

0 1 0 1 0 1 0 1

0 1 0 1 0 1 0 1